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Path tracking control of a self-driving wheel excavator via an enhanced data-driven model-free adaptive control approach

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Abstract: In this work, an enhanced model-free adaptive control algorithm considering the time delay (EMFAC-TD) is proposed for a class of multi-input and multi-output systems and is further applied to the path tracking control problem of a self-driving wheel excavator. First, a preview-deviation-yaw-based method is first proposed for the simplification of excavator dynamics. Then, based on the simplified dynamics, the EMFAC-TD controller is designed via a novel dynamical linearisation technique with a time-varying parameter termed pseudo Jacobian matrix, which contains the coupling information of the excavator driving dynamics. Moreover, for the time delay of the excavator, EMFAC-TD method also handles it by introducing the Smith estimating method. The main feature of the proposed EMFAC-TD is that the controller is designed only based on the input and output data of the excavator. Further, the stability of the proposed EMFAC-TD is proved via rigorous mathematical analysis. The experimental results implemented on a real self-driving excavator show that the maximum tracking error of the excavator controlled via EMFAC-TD method can be limited as 0.3–0.7 m for different driving tasks, satisfying the practical requirement. The experimental results verify that EMFAC-TD method can improve the tracking performance and complete the required task successfully.

1 Introduction

Excavators are employed in many industrial fields, including agriculture, construction and military applications [1]. Generally, excavator vehicles are controlled manually, where a human operator plans with a driver. However, when performing repetitive or dangerous tasks in hazardous environments, a self-driving excavator becomes beneficial.

For the control of excavators, there have already been many studies. Saeedi *et al.* [2] describe a vision-based control system for an excavator. The system includes several controllers that collaborate to move a mobile vehicle from a starting position to a goal position. In [3], a novel approach was presented for the position-based impedance control of excavator arms. Bender *et al.* [4] propose a modelling method for a hydraulic mini-excavator, and then, offset-free model predictive controllers (MPCs) are proposed and applied based on the established mathematical model. Abhari *et al.* [5] present an adaptive robust control algorithm for the non-linear dynamics of robot manipulators with unknown backlash in gears based on the basic non-linear model of a serial manipulator robot. Andaluz *et al.* [6] present a unified motion control scheme for the point stabilisation and trajectory tracking problems of quadcopters based on the kinematic model of the unmanned aerial vehicles, and the stability and robustness of the proposed method are proved by the Lyapunov's method. Wang *et al.* [7] investigate the dynamic of mechanical structure considering all main parameter uncertainties and disturbances and propose a novel robust feedback tracking controller considering both the optimisation of path tracking and the minimisation of the power consumption energy is proposed.

The above studies [2–7] employed model-based control approaches that require a mathematical representation of the target plant. However, when an excavator is driving, its dynamics are a complex system and usually change with different road conditions and driving tasks. Thus, it is difficult to establish a precise mathematical model of the excavator dynamics, which means that those model-based control methods are difficult to apply.

With the developments of software, hardware and the calculating ability of computers, large amounts of driving data can be collected online during the driving process of the excavator. Therefore, data-driven control (DDC) methods have become a reasonable manner by which to address the control problem of the self-driving excavator [8].

There exist many different types of data driving control methods, e.g. Campi and Savaresi [9] proposed a virtual reference feedback tuning (VRFT) approach, in which the controller can be constructed using the offline input and output data by introducing a termed virtual reference single concept. The authors of [10, 11] propose a series of iterative-based methods, in which the controller can be offline obtained via an iterative learning manner. Moreover, the authors of [12–16] propose a series of adaptive dynamical programming control methods combined the conventional optimal control theories. Besides, the authors of [17, 18] present a subspace identification method by combining the idea of model predictive control (MPC) method. The abovementioned DDC methods mainly design controllers via s offline manner. Besides, there also exist some online DDC methods describing as follows. The authors of [19–21] propose a series of model free control methods based on conventional PID method and the differential algebra. In addition, the authors of [22, 23] propose a kind of fuzzy control methods, in which the fuzzy logic idea is introduced into the controller design. Besides, Bontempi *et al.* [24] present a lazy learning-based controller design method by using the local online learning idea.

As a representative DDC algorithm, the model-free adaptive control (MFAC) method [25, 26] has developed rapidly in recent years. For a class of non-linear discrete-time systems, the essential idea of MFAC is that an equivalent dynamic linearisation technique is first used at each current operating point with a novel concept

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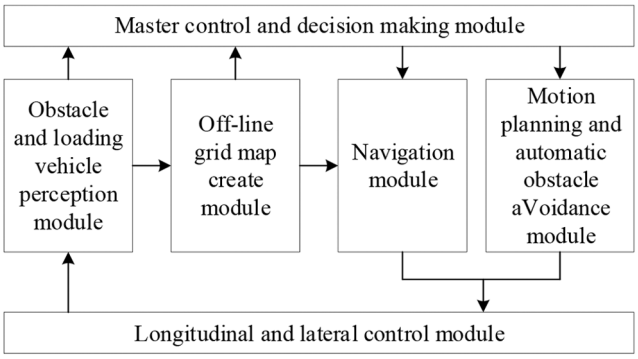


Fig. 1 *Design frame for the self-driving excavator*

termed pseudo gradients (PGs, for single-input and single-output systems) or a pseudo Jacobian matrix (PJM, for multi-input and multi-output (MIMO) systems), by which the current discrete-time non-linear system is replaced by this linear data model, whose parameters are estimated online using the input and output data of the system. In the sequel, the MFAC controller is designed via the data model.

Compared with the existing DDC methods, the advantages of MFAC are that it is adaptive, easy to implement, and has theoretical guarantees of the closed-loop stability and convergence. So far, MFAC has been further researched and been widely applied successfully, such as in a wide power system [27], the Wood-Berry distillation column [28] artificial control [29], the concentration control of the enriched oxygen in the synthetic ammonia industry [30], a high-speed senseless BLDC motor [31], the path tracking of an autonomous car [32, 33], and even in image processing [34].

To date, MFAC has been investigated from many theoretical perspectives. Hou and Jin [35] propose an MFAC algorithm for a class of discrete MIMO systems. Jin and Hou [36] propose a novel MFAC algorithm considering the time delay. However, as far as the authors know, for MIMO systems considering the time delay, there have been no studies of MFAC for this problem. Additionally, the self-driving excavator investigated here is a non-linear MIMO system, and it has a large time delay due to the inertance caused by its large mass. Therefore, it is necessary to investigate MFAC theories considering both the MIMO system and the system time delay.

In this paper, a novel enhanced data-driven model-free adaptive control algorithm considering the time delay (EMFAC-TD) is proposed for a class of MIMO systems, and then the method is applied in the path tracking control of a self-driving wheel excavator. The main feature of proposed EMFAC-TD is that the controller is designed in a data-driven manner and only based on the input and output data of the excavator using a new dynamic linearisation technique with a time-varying parameter named the PJM. Moreover, the method also considers the time delay of the system by using the Smith estimating method and the tracking differentiators. Further, the stability of the proposed EMFAC-TD is proved via rigorous mathematical analysis.

For the application field, the dynamics of the excavator are a complex MIMO system with a time delay, and there exists a strong coupling between the longitudinal driving speed and the lateral wheel angle. For these reasons, a preview-deviation-yaw-based method is first proposed for the formulation of the driving dynamics of the excavator, by which the excavator dynamics is simplified. Then, the proposed EMFAC-TD is successfully applied in the control process. The experimental results, implemented on a real self-driving excavator in Xuzhou, Jiangsu Province, show that EMFAC-TD can complete the required task and solve the control problem successfully.

The outline of this paper is organised as follows. Section 2 introduces the control platform of the self-driving excavator, proposes a preview-deviation-yaw-based formulation method for the lateral dynamics of the self-driving excavator, and then formulates the dynamics of the excavator. Section 3 proposes the EMFAC-TD control algorithm and presents the stability analysis and implementation issues on EMFAC-TD. The experimental results for a self-driving excavator are shown in Section 4, and the conclusion is provided in Section 5.

2 Problem formulation

2.1 Overall self-driving process of excavator

The whole operating process of the investigated excavator is shown in Fig. 1, and each module in Fig. 1 is introduced in Section 2.1.1.

2.1.1 Introduction of the control and decision units: The control and decision unit, including the human–computer interaction interface and the native autonomous driving task setting programme, is the core module of the self-driving excavator and the central unit of the control and information processing of other modules in the system. The decision module is the central unit of the self-driving wheel excavation. It directly controls the unified deployment of the states and commands of the global driving path, obstacle detection and tracking, obstacle avoidance, navigation, speed and steering.

Navigation unit: The navigation module is one of the main units of the self-driving wheeled excavator. Specifically, it contains a core component that is responsible for receiving the original information of the body movement collected by the inertial navigation equipment, and it also performs the self-driving wheel excavation based on stored map information. Consequently, the self-driving wheel can drive safely on the predefined route.

Detection and tracking unit for obstacles and loading vehicles: The environment perception unit is the core module group, by which the self-driving wheeled excavator system perceives the surrounding operations and driving environment. It not only provides the basis for the decision-making process of the superior master control and decision unit, but also serves as the basis for the operation of the motion planning algorithm.

Motion planning and autonomous obstacle avoidance unit: As one of the cores of the self-driving wheeled excavator system, the planning system determines the desired motion parameters of the platform under various constraints, as well as path planning. These constraints include environmental constraints (such as dynamic and static obstacles around the platform), guidance constraints (such as the target pose of the movement), platform kinematic constraints, and dynamic constraints.

Longitudinal control and lateral control units: The longitudinal and lateral control units are the core modules of the self-driving wheeled excavator body control. The main function of these modules is to control the driving speed and steering wheel angle of the excavator by controlling corresponding actuators.

The vehicle speed control system is also called the longitudinal control system. By adjusting the accelerator pedal and the brake pedal, the actual speed can be controlled and provided by the IMU or the transmission speed sensor. The direction control system is known as the lateral control system. The internal small closed loop consists of a steering controller and a steering wheel steering angle sensor, and the controller parameters are set by the manufacturer. The external large closed loop consists of the wheel angle and the direction controller.

In addition to the abovementioned modulus, the concepts of the look-ahead point and look-ahead distance will be introduced before the control for a self-driving excavator, which will be described in Section 2.1.2.

2.1.2 Look-ahead point and look-ahead distance: When the excavator is driving, the driver usually focuses on an imaginary 'target point' in front of the excavator. This point is named the look-ahead point. Even when the excavator has no driver, this look-ahead point is still needed, and it is equivalent to the time-varying desired signal for a control system. The distance between the excavator and the look-ahead point is named the look-ahead distance. A prototype of the excavator and an abridged general view of the look-ahead point and look-ahead distance are shown in Fig. 2.

In most cases, the look-ahead distance is calculated via an empirical equation

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$$L=\begin{cases}0, & v=0\\ l_{\min}, & 0<v\le v_{\min}\\ a_0v+l_{\min}, & v_{\min}<v\le v_{\mathrm{mid}},\\ a_1v^2+a_0v+l_{\min}, & v_{\mathrm{mid}}<v\le v_{\max}\\ l_{\max}, & v>v_{\max}\end{cases}\tag{1}$$

where $l_{\max}$ and $l_{\min}$ are the minimum, the maximum and the distances between the excavator and the look-ahead point, which are related to the perception ability of the devices in the excavator. When the excavator stops, the look-ahead distance is 0. Moreover, for most cases, when the excavator runs with a low speed within the interval $(0,\ v_{\min}]$, the look-ahead distance can be considered as a constant $l_{\min}$. Moreover, when the excavator runs with a larger speed in the interval $[v_{\min},\ v_{\mathrm{mid}}]$, the look-ahead distance becomes a linear relationship with the driving speed. Besides, for very few special cases, the excavator running with a larger speed in the interval $[v_{\mathrm{middle}},\ v_{\max}]$, the look-ahead distance will have a quadratic relationship with the driving speed. In addition, when the excavator runs larger than $v_{\max}$, due to the perception ability of the devices, the look-ahead distance of the excavator is limited as a constant $l_{\max}$. By performing large number of practical tests on the devices, the parameters of (1) are determined as follows: $l_{\min}=4$ m, $l_{\max}=27$ m, $v_{\min}=5$ km/h, $v_{\mathrm{mid}}=12$ km/h, $v_{\mathrm{mid}}=18$ km/h, $v_{\mathrm{mid}}=18$, $a_0=1.12$, $a_1=0.85$.

2.2 Control system formulation

From the introduction of the longitudinal control and lateral control units mentioned in Section 2.2.1, we know that the control problem of the self-driving excavator can be analysed from the lateral and longitudinal directions. Specifically, the control system structure is shown in Fig. 3.

From Fig. 3, it can be seen that during the driving process, the equipped sensors continuously transmit the excavator state information data to the host computer, by which the desired speed and desired position can be planned. Then, the objective of the control system is to make the measured speed and the measured position accurately track the desired speed and the position by adjusting the angle of the steering wheel and the output of the pressure on the throttle or the brake pedals.

2.2.1 Longitudinal dynamics: For the longitudinal direction, the control objective is the driving speed, which can be controlled by using the opening degree of the throttle and brake pedals. The driving speed is also influenced by other factors, such as the wind speed, the road conditions, and the angle of the steering wheel. Consequently, the dynamics of the driving speed of the excavator can be described as follows:

$$\begin{aligned}y_1(k+1)=f_1(u_1(k-\tau),\ldots,u_1(k-\tau-n_u+1),\\ y_1(k),\ldots,y_1(k-n_y),w_1(k)),\end{aligned}\tag{2}$$

where y_1 denotes the driving speed of the excavator, u_1 denotes the opening degree of the throttle and brake pedals, and n_y and n_u are unknown positive constants. τ is the time delay of the system. $w_1(k)$ denotes other unmodelled dynamics that affect the longitudinal dynamics.

2.2.2 Lateral dynamics: The lateral control process is fairly important, because it directly determines the path tracking performance of the self-driving excavator. We must control the driving direction of the excavator under any situation, including different driving speed and road conditions. The lateral dynamics of the excavator are described in Fig. 4*a*.

In Fig. 4, a yaw angle α is first defined, which is the included angle between the north direction and the driving direction of the excavator. Then, the coordinate diagram when the vehicle is driving can be shown in Fig. 4*a*.

Moreover, a proposed angle θ is defined as the included angle between the driving direction and the direction from the current position to the look-ahead point, named the pre-aim-yaw, which is shown in Fig. 4*b*. For the connection of the defined θ, AB, α, α^* a following theorem is given.

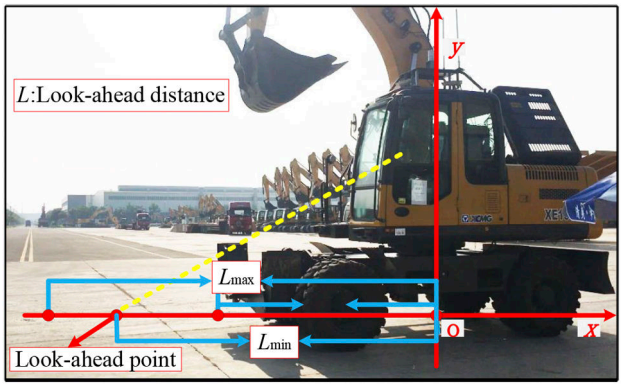


Fig. 2 *Abridged general view of the look-ahead point and look-ahead distance*

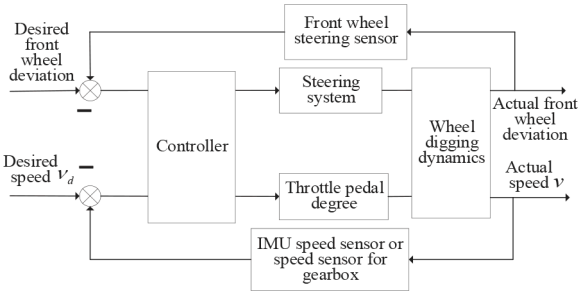


Fig. 3 *Control system structure of the self-driving excavator*

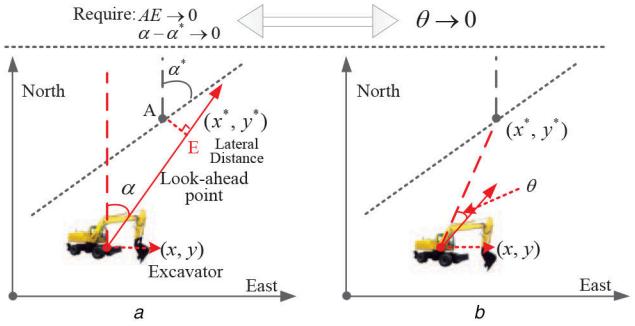


Fig. 4 *Dynamics formulation of lateral control process*
(a) The coordinate diagram when the vehicle is driving, ***(b)*** The coordinate diagram when the vehicle is driving describing via pre-aim-yaw

Theorem 1: If θ, AE, α, α^* are defined in Fig. 4, then the following conclusion can be given

$$\theta\to 0\Leftrightarrow(x,y)\to(x^*,y^*),\tag{3}$$

where (x,y) are the coordinates of the current excavator, and (x^*, y^*) are denoted as the coordinates of the look-ahead point.

Proof: See Appendix 1. □

The pre-aim-yaw θ can be controlled by the angle of the steering wheel. In the sequel, θ is renamed y_2.

Specifically, the relationship between the tracking situation of $y_2(k)$ and the operational situation of the excavator can be observed in Figs. 5 and 6.

From Figs. 5 and 6, it can be concluded that the controller design for the lateral control system is important. If $y_2(k)$ is controlled without overshoot, then the self-driving excavator will be controlled well, i.e. the conclusion of Theorem 1.

As a result, the dynamics of the lateral control process can be formulated as follows:

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$$y_2(k+1) = f_2(u_2(k-\tau), \ldots, u_2(k-\tau-n_u+1), y_2(k), \ldots, y_2(k-n_y), w_2(k)), \tag{4}$$

where u_2 denotes the angle of the steering wheel. $w_2(k)$ denotes other unmodelled dynamics that affect the lateral dynamics. For the lateral dynamics (4), the control objective is to calculate an optimal $u_2(k)$ such that $y_2(k)$ can track the zero without overshoot as fast as possible.

2.2.3 Dynamic formulation and control difficulties of the excavator: Considering the dynamics of the longitudinal and lateral directions, it can be concluded that during the driving process, the variables of the excavator that need to be controlled are the driving speed $y_1(k)$ and the defined pre-aim-yaw angle $y_2(k)$, which can be adjusted via the opening degree of the throttle and brake pedals $u_1(k)$ and the angle of the steering wheel $u_2(k)$. Consequently, the control system of the excavator can be expressed as a two-input and two-output non-linear system as follows:

$$\boldsymbol{y}(k+1) = \boldsymbol{f}(\boldsymbol{y}(k), \ldots, \boldsymbol{y}(k-n_y), \boldsymbol{u}(k-\tau), \ldots, \boldsymbol{u}(k-n_u-\tau)), \tag{5}$$

where $\boldsymbol{u}(k) = [u_1(k),\ u_2(k)]^{\mathrm{T}}$, $\boldsymbol{y}(k) = [y_1(k),\ y_2(k)]^{\mathrm{T}}$, and $\boldsymbol{f}(\cdot)$ is an unknown non-linear function.

From (5), the control difficulties of the excavator are as follows.

Strong coupling: First, there exists a strong coupling between the control inputs and the control outputs. For example, when the excavator completes a turning task, both the driving speed and the angle of the steering wheel can influence the turning performance. Specifically, for a given turning task, a faster speed of the excavator is more difficult to achieve than a slower speed. Therefore, how to deal with the existing strong coupling is a considerable problem for the controller design.

Difficult to model: Due to the large inertance, high non-linearity, strong coupling and large time delay of the wheel excavation body dynamics, high dynamic quality and high-precision directional control systems are difficult to achieve, which means that a conventional model-based control algorithm is not sufficient. In addition, the control dynamics of the excavator usually change during the driving process, since the excavator will meet different road conditions and difficult driving environments. For these reasons, the designed controller should be adaptive and able to update the controller parameters in real time. The controller should also be designed without the information of the precise mathematical model of the system.

Strong inertia and time delay: The excavator has a large weight, which gives the control process a time delay. Therefore, during the controller design process, how to effectively deal with the time delay is also a considerable problem.

3 EMFAC-TD controller design

Considering the abovementioned factors, this section will propose a novel model-free adaptive control method. The algorithm can be described using the following parts.

3.1 Dynamic linearisation

For the system (5), the following assumptions are first made.

Assumption 1: The partial derivative of $\boldsymbol{f}(\cdot)$ with respect to the control input $\boldsymbol{u}(k)$ is continuous.

Assumption 2: The investigated system (5) is generalised Lipschitz, i.e. for any $k_1 \neq k_2$, $k_1, k_2 \geq 0$ and $\boldsymbol{u}(k_1) \neq \boldsymbol{u}(k_2)$, it has $\| \Delta \boldsymbol{y}(k+1) \| \leq C \| \Delta \boldsymbol{u}(k) \|$ for any k and $\Delta u(k) \neq 0$ with $\Delta \boldsymbol{y}(k+1) = \boldsymbol{y}(k+1) - \boldsymbol{y}(k)$, $\Delta \boldsymbol{u}(k) = \boldsymbol{u}(k) - \boldsymbol{u}(k-1)$ and C is a positive constant.

Remark 1: In practice, these assumptions are reasonable and acceptable. Specifically, Assumption 1 means that the speed and the pre-aim-yaw excavator continuously change with the control inputs, which is reasonable during the driving process. Moreover, Assumption 2 means that system outputs of the excavator cannot go to infinity if the rates of variation of the system inputs are bounded. In other words, if the excavator is given a finite opening degree of the throttle and finite angle of the steering wheel, the corresponding speed and turning angle are also limited.

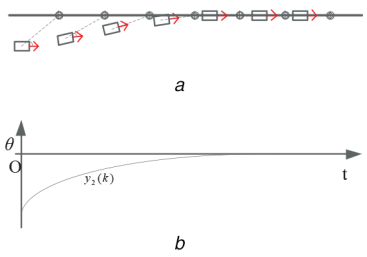


Fig. 5 *Relationship of the excavator and $y_2(k)$ without overshoot*
(a) Operational situation of the self-driving excavator, ***(b)*** Tracking situation of $y_2(k)$

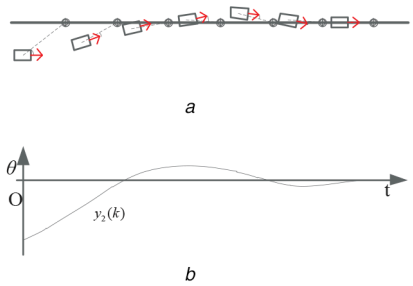


Fig. 6 *Relationship of the excavator and $y_2(k)$ with overshoot*
(a) Operational situation of the self-driving excavator, ***(b)*** Tracking situation of $y_2(k)$

In the sequel, the following theorem is introduced and proven as follows.

Theorem 2: For system (5) satisfying Assumptions 1 and 2, there exists a $\boldsymbol{\Phi}(k) \in \boldsymbol{R}^{2\times 2}$ with elements $\phi_{ij}(k)$, $i = 1, 2$, $j = 1, 2$, called the pseudo Jacobian matrix (PJM), such that if $\| \Delta \boldsymbol{u}(k) \| \neq 0$, system (5) can be transformed into the following model:

$$\boldsymbol{y}(k+\tau+1) = \boldsymbol{y}(k+\tau) + \boldsymbol{\Phi}(k)\Delta \boldsymbol{u}(k), \tag{6}$$

and for any time k, $\boldsymbol{\Phi}(k)$ is bounded, denoted as

$$\boldsymbol{\Phi}(k) = \begin{bmatrix} \varphi_{11}(k) & \varphi_{12}(k) \\ \varphi_{21}(k) & \varphi_{22}(k) \end{bmatrix}.$$

Proof: See Appendix 2. □

For a rigorous analysis of the stability, another assumption is given.

Assumption 3: The PJM $\boldsymbol{\Phi}(k)$ is a diagonally dominant matrix in the following sense: $|\varphi_{ij}(k)| < b_1$, $b_2 \leq |\varphi_{ii}(k)| \leq \alpha b_2$, $\alpha \geq 1$, $b_2 > b_1(2\alpha+1)(p-1)$, $i = 1, 2$, $j = 1, 2$, $i \neq j$, and the signs of all the elements of $\boldsymbol{\Phi}(k)$ are fixed.

Remark 2: Assumption 3 describes the relationship between the input and output data of the closed-loop system. Since the plant model is unavailable and only the system operation data can be used, the diagonal dominance condition of the system I/O data relationship may be the last choice to describe system coupling. Moreover, the condition on the fixed sign of all the elements of $\boldsymbol{\Phi}(k)$ is similar to the control direction in the model-based control methods. For instance, in most of the existing adaptive control methods, the control gain is usually assumed to be constant with a known sign. Roughly speaking, this condition could be verified if the I/O data of the controlled system are rich and accurate enough.

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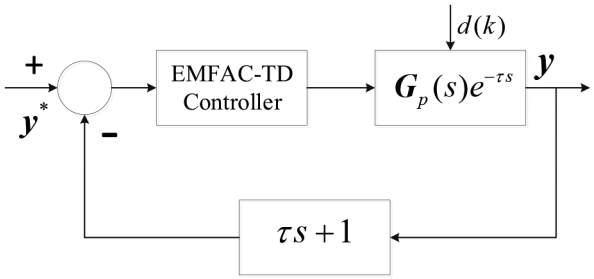


Fig. 7 *Block diagram of EMFAC-TD control system*

3.2 Control system design

3.2.1 Tackling the time delay: In (6), $\boldsymbol{y}(k+\tau)$ is unknown at time instant k. To address this problem, the Smith estimation method is introduced here. The structure of the control system is shown in Fig. 7, in which the transfer function of the investigated system is $\boldsymbol{G}_p(s)e^{-\tau s}$ with a time delay τ. In the feedback part, a leading part can be utilised to predict the further system output. When the plant is slowly time-varying, $e^{-\tau s}$ can be replaced by $\tau s+1$, which can complete the function of the Smith estimation. Thus, the dynamic linearisation model can be further converted to

$$\boldsymbol{y}(k+\tau+1)=\boldsymbol{y}(k)+\tau\boldsymbol{y}'(k)+\boldsymbol{\Phi}(k)\Delta\boldsymbol{u}(k), \tag{7}$$

where $\boldsymbol{y}'(k)$ is the derivative vector of the output $\boldsymbol{y}(k)$.

Remark 3: This work mainly investigates the situation that the time delay τ is a constant. In some practical cases, the time delay of the investigated system is unknown. Thus, it can be assumed that the time delay is bounded, i.e. $\tau(k)\in[\tau_{\min},\tau_{\max}]$, where $\tau_{\min}$ and $\tau_{\max}$ denote the lower bound and the upper bound of the time delay, respectively. Meanwhile, it also can be assumed that the time delay varies with the time instant k, described as

$$\tau(k)=g(u(k),y(k),u(t-\tau(k)),\ k,d(k)), \tag{8}$$

where $d(k)$ denotes other uncertain factors that can affect the changing of $\tau(k)$. The function $g(\cdot)$ can be divided into following two cases. *Case 1:* $g(\cdot)$ is known, or $g(\cdot)$ can be estimated as $\hat{g}(\cdot)$, which means that the changing of $\tau(k)$ is known or can be estimated as $\hat{\tau}(k)$ at each time instant. In this case, the original τ in the control scheme can be repaced as $\tau(k)$, or $\hat{\tau}(k)$. *Case 2:* $g(\cdot)$ is unknown, which means that in each time instant k, the time delay τ is changing and unknown. In this case, we consider the worst situation, i.e. let $\tau_{\max}$ replace $\tau(k)$ at each time instant k.

3.2.2 Controller structure design: For the control law, a weighted one-step-ahead control input cost function is adopted and given by

$$\begin{aligned}J_1(\boldsymbol{u}(k))=&\parallel \boldsymbol{y}^*(k+\tau+1)-\boldsymbol{y}(k+\tau+1)\parallel^2\\&+\lambda\parallel\boldsymbol{u}(k)-\boldsymbol{u}(k-1)\parallel^2\end{aligned}, \tag{9}$$

where λ is a weighting constant, which is introduced to restrict the changing rate of the control input. Meanwhile, it is often used in control system design to assure the smoothness of the control input signal. In fact, λ is an important adjustable parameter for the control system design. Theoretical analysis and experiments also show that proper selection of λ can guarantee the stability and good output tracking performance of the controlled system. Generally, the λ can be selected according to the nature of the control systems. For the slow time-varying systems, in which the changes of system inputs are small, the λ can be chosen as a small interval, e.g. [0.5, 5]. Moreover, for other complex systems, in which the changes of system input changes dramatically, the λ can be appropriately increased to restrain the drastic changes of control inputs.

Substituting (7) into (9) with respect to $\boldsymbol{u}(k)$. Let the left side of the equation be zero, it can be inferred that

$$\begin{aligned}\boldsymbol{u}(k)=&\boldsymbol{u}(k-1)+(\lambda\mathbf{I}+\boldsymbol{\Phi}^T(k)\boldsymbol{\Phi}(k))^{-1}\boldsymbol{\Phi}^T(k)\\&\times(\boldsymbol{y}^*(k+\tau+1)-\tau\boldsymbol{y}'(k)-\boldsymbol{y}(k)).\end{aligned} \tag{10}$$

It is noted that the control algorithm (10) requires calculating the matrix inversion, which is very difficult when the dimensions of the system input and output are large. Therefore, the control algorithm (10) is not suitable for practical applications. To solve this problem, a simplified control algorithm is given as follows:

$$\boldsymbol{u}(k)=\boldsymbol{u}(k-1)+\frac{\rho\boldsymbol{\Phi}^T(k)(\boldsymbol{y}^*(k+\tau+1)-\tau\boldsymbol{y}'(k)-\boldsymbol{y}(k))}{\lambda+\parallel\boldsymbol{\Phi}(k)\parallel^2}, \tag{11}$$

where the step factor $\rho\in(0,1]$ is introduced to make the control algorithm more general, and it will be used in the stability analysis.

Remark 4: In order to avoid the calculation of matrix inversion, the controller scheme (11) is obtained via a simplified version. Moreover, Theorem 3 verifies the stability and the bounded input bounded output (BIBO) property of the control scheme.

In (11), the PJM $\boldsymbol{\varphi}(k)$ is unknown and time-varying, and its estimated value $\hat{\boldsymbol{\varphi}}(k)$ is utilised here to replace $\boldsymbol{\varphi}(k)$. For the estimation of $\boldsymbol{\varphi}(k)$, there exist some typical methods, such as the modified projection algorithm [34], the least-squares algorithm with a time-varying forgetting factor [18], and the local learning algorithm [27, 35]. In this paper, we use the modified projection algorithm to estimate the unknown PJM. Specifically, a cost function of the PJM estimation is given as

$$\begin{aligned}J_2(\boldsymbol{\Phi}(k))=&\parallel\Delta\boldsymbol{y}(k)-\boldsymbol{\Phi}(k-\tau-1)\Delta\boldsymbol{u}(k-\tau-1)\parallel^2\\&+\mu\parallel\boldsymbol{\Phi}(k)-\boldsymbol{\Phi}(k-\tau-1)\parallel^2,\end{aligned} \tag{12}$$

where the weighting factor $\mu>0$ is used to restrain the change in the PJM estimation.

Letting $\partial J_2/\partial\boldsymbol{\Phi}(k)=0$, we have the following projection algorithm:

$$\begin{aligned}\hat{\boldsymbol{\Phi}}(k)=&\hat{\boldsymbol{\Phi}}(k-\tau-1)+(\Delta y(k)-\hat{\boldsymbol{\Phi}}(k-\tau-1)\\&\times\Delta\boldsymbol{u}(k-\tau-1))\Delta\boldsymbol{u}^T(k-\tau-1)\\&\times(\mu I+\Delta\boldsymbol{u}(k-\tau-1)\Delta\boldsymbol{u}^T(k-\tau-1))^{-1}.\end{aligned} \tag{13}$$

Since the estimation algorithm (13) contains matrix inversion, the following simplified PJM estimation algorithm is used:

$$\begin{aligned}\hat{\boldsymbol{\Phi}}(k)=&\hat{\boldsymbol{\Phi}}(k-\tau-1)\\&+\frac{\eta(\Delta\boldsymbol{y}(k)-\hat{\boldsymbol{\Phi}}(k-\tau-1)\Delta\boldsymbol{u}(k-1))\Delta\boldsymbol{u}^T(k-1)}{\mu+\parallel\Delta\boldsymbol{u}(k-1)\parallel^2},\end{aligned} \tag{14}$$

where $\eta\in(0,2)$ is a weighting factor and

$$\hat{\boldsymbol{\Phi}}(k)=\begin{bmatrix}\hat{\varphi}_{11}(k) & \hat{\varphi}_{12}(k)\\ \hat{\varphi}_{21}(k) & \hat{\varphi}_{22}(k)\end{bmatrix}\in R^{2\times2}$$

is the estimation of the unknown PJM $\boldsymbol{\Phi}(k)$.

In order to make the estimation algorithm have a stronger ability in tracking time-varying parameter, the following reset algorithm (15) and (16) is added. Remarkably, this reset mechanism does not occur frequently in most practical cases. Only when the control process becomes abnormal, the mechanism can be triggered as a last resort. e.g. a sudden disturbance or abnormal behaviour for the system

$$\begin{aligned}&\hat{\varphi}_{ij}(k)=\hat{\varphi}_{ij}(1),\ \text{if}\ |\hat{\varphi}_{ij}(k)|>b_1\ \text{or}\\&\operatorname{sign}(\hat{\varphi}_{ij}(k))\neq\operatorname{sign}(\hat{\varphi}_{ij}(1)),\quad i,j=1,2,\ i\neq j,\end{aligned} \tag{15}$$

$$\begin{aligned}&\hat{\varphi}_{ij}(k)=\hat{\varphi}_{ij}(1),\ \text{if}\ |\hat{\varphi}_{ij}(k)|<b_2\ \text{or}\ |\hat{\varphi}_{ij}(k)|>\gamma b_2\\&\operatorname{sign}(\hat{\varphi}_{ii}(k))\neq\operatorname{sign}(\hat{\varphi}_{ii}(1)),\quad i=1,2\end{aligned} \tag{16}$$

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3.2.3 Tracking differentiator: In practice, the controller (11) has the following two problems. One problem is that $\boldsymbol{y}'(k)$ cannot be achieved in a physical manner. In general, $\boldsymbol{y}'(k)$ is replaced by the first-order difference $(\boldsymbol{y}(k)-\boldsymbol{y}(k-1))/T$ in most cases, where T is the sampling period. However, when $\boldsymbol{y}(k)$ is polluted by noise, $(\boldsymbol{y}(k)-\boldsymbol{y}(k-1))/T$ will be seriously polluted by $n(k)/T$, where $n(k)$ denotes the noise at time instant k.

To deal with this problem and obtain $\boldsymbol{y}'(k)$, the tracking differentiator [36] is introduced here. These two tracking differentiators are described as follows for $y_1'(k)$ and $y_2'(k)$ in $\boldsymbol{y}'(k)$.

Tracking differentiator 1 (for $y_1'(k)$):

$$\begin{cases} y_{11}(k+1)=y_{11}(k)+h_{11}y_{12}(k), \\ y_{12}(k+1)=y_{12}(k) \\ +h_{11}\cdot fhan(y_{11}(k)-y_1(k),\ y_{12}(k),\ r_1,\ h_{12}) \end{cases}, \tag{17}$$

Tracking differentiator 2 (for $y_2'(k)$):

$$\begin{cases} y_{21}(k+1)=y_{21}(k)+h_{21}y_{22}(k), \\ y_{22}(k+1)=y_{22}(k) \\ +h_{21}\cdot fhan(y_{21}(k)-y_2(k),\ y_{22}(k),\ r_2,\ h_{22}) \end{cases}, \tag{18}$$

where h_{11}, h_{21} are the sampling periods and r_1, r_2, h_{12}, h_{22} are adjustment parameters. In most cases, h_{11} and h_{21} are set equal to h_{12} and h_{22}. $y_{11}(k)$ and $y_{12}(k)$ track $y_1(k)$ and $y_1'(k)$, respectively, and $y_{21}(k)$ and $y_{22}(k)$ track $y_2(k)$ and $y_2'(k)$, respectively, [36].

In addition, $fhan(x_1,x_2,r,h_0)$ in (17) and (18) is defined as follows:

$$\begin{cases} f=h_0r^2,\ \ b_0=h_0x_2, \\ \gamma=x_1+b_0, b_1=\sqrt{f(f+8|\gamma|)} \\ b_2=b_0+\mathrm{sign}(\gamma)(b_1-f)/2 \\ s_\gamma=(\mathrm{sign}(\gamma+f)-\mathrm{sign}(\gamma+f))/2 \\ b=(b_0+\gamma-b_2)s_\gamma+b_2 \\ s_b=(\mathrm{sign}(b+f)-\mathrm{sign}(b+f))/2 \\ fhan=-r(b/f-\mathrm{sign}(b))s_b-r\mathrm{sign}(a) \end{cases}. \tag{19}$$

Notably, (17) and (18) are time-optimal solutions that guarantee the fastest convergence from $y_{11}(k)$ and $y_{21}(k)$ to $y_1(k)$ and $y_2(k)$ without any overshoot, respectively. $y_{12}(k)$ and $y_{22}(k)$ can also be smoothed via $y_1'(k)$ and $y_2'(k)$, respectively [36].

By using (17)–(19), the abovementioned problems are solved as follows: (1) $y_1'(k)$ can be replaced by $y_{12}(k)$; (2) $y_2'(k)$ is transited via $y_{22}(k)$. Therefore, denote $y_{\mathrm{TD}}'=[y_{12},\ y_{22}]^{\mathrm{T}}$, and then the controller (11) can be written as

$$\boldsymbol{u}(k)=\boldsymbol{u}(k-1)+\frac{\rho\boldsymbol{\Phi}^T(k)(\boldsymbol{y}^*(k+\tau+1)-\tau\boldsymbol{y}_{\mathrm{TD}}'(k)-y(k))}{\lambda+\|\ \boldsymbol{\Phi}(k)\ \|^2}. \tag{20}$$

Remark 5: During the driving process of the excavator, the lateral dynamics and the longitudinal dynamics are coupled with each other. The longitudinal control process focuses on driving speed, whereas the lateral control process focuses on the driving direction. The driving speed can affect the control of the driving direction. For example, when the excavator makes a quarter turn, the driving direction control with a high driving speed is more difficult than that of low driving speed. Meanwhile, when the excavator is driving, the change of driving direction also affects the control of the driving speed. For this problem, the controller introduces a time-varying parameter named the pseudo Jacobian matrix (PJM), which includes the coupling dynamics of the excavator at each operating point. Moreover, the PJM is updated online, so that the system dynamics can be described in real time.

3.3 System control scheme and stability analysis

3.3.1 System control scheme: By combining the PJM estimation algorithm (14) and control algorithm (18), the EMFAC-TD scheme of discrete-time MIMO non-linear systems is constructed as follows:

$$\hat{\boldsymbol{\Phi}}(k)=\hat{\boldsymbol{\Phi}}(k-\tau-1)+\frac{\eta(\Delta\boldsymbol{y}(k)-\hat{\boldsymbol{\Phi}}(k-\tau-1)\Delta\boldsymbol{u}(k-\tau-1))\Delta\boldsymbol{u}^T(k-\tau-1)}{\mu+\|\ \Delta\boldsymbol{u}(k-\tau-1)\ \|^2}, \tag{21}$$

$$\hat{\varphi}_{ij}(k)=\hat{\varphi}_{ij}(1),\ \text{if}\ |\hat{\varphi}_{ij}(k)|>b_1\ \text{or}\ \mathrm{sign}(\hat{\varphi}_{ij}(k))\neq\mathrm{sign}(\hat{\varphi}_{ij}(1)),\ i,j=1,2,\ i\neq j, \tag{22}$$

$$\hat{\varphi}_{ij}(k)=\hat{\varphi}_{ij}(1),\ \text{if}\ |\hat{\varphi}_{ij}(k)|<b_2\ \text{or}\ |\hat{\varphi}_{ij}(k)|>\gamma b_2\ \mathrm{sign}(\hat{\varphi}_{ii}(k))\neq\mathrm{sign}(\hat{\varphi}_{ii}(1)),\ i=1,2 \tag{23}$$

$$\boldsymbol{u}(k)=\boldsymbol{u}(k-1)+\frac{\rho\boldsymbol{\Phi}^T(k)(\boldsymbol{y}^*(k+\tau+1)-\tau\boldsymbol{y}_{\mathrm{TD}}'(k)-y(k))}{\lambda+\|\ \boldsymbol{\Phi}(k)\ \|^2}, \tag{24}$$

$$\boldsymbol{y}_{\mathrm{TD}}'=[y_{12},\ y_{22}]^{\mathrm{T}}$$

$$\begin{cases} y_{11}(k+1)=y_{11}(k)+h_{11}y_{12}(k), \\ y_{12}(k+1)=y_{12}(k) \\ +h_{11}\cdot fhan(y_{11}(k)-y_1(k),\ y_{12}(k),\ r_1,\ h_{12}) \end{cases}, \tag{25}$$

$$\begin{cases} y_{21}(k+1)=y_{21}(k)+h_{21}y_{22}(k), \\ y_{22}(k+1)=y_{22}(k) \\ +h_{21}\cdot fhan(y_{21}(k)-y_2(k),\ y_{22}(k),\ r_2,\ h_{22}) \end{cases}, \tag{26}$$

where $\hat{\varphi}_{ij}(1)$ is the initial value of $\hat{\varphi}_{ij}(k)$, $i=1,2,j=1,2$, $\lambda>0$, $u>0$, $\eta\in(0,2)$ and $\rho\in(0,1]$. $fhan(\cdot)$ is defined in (17).

3.3.2 Stability analysis: To make the controller more general, the proof extends the dimension of the control input and control output from 2 to m, respectively, i.e. $i=1,2,\ldots,m$, $j=1,2,\ldots,m$

The following lemma and assumption will be used for the stability analysis.

Lemma 1: Let $M=(m_{ij})\in C^{n\times n}$. For each $1\le i\le n$, the Gerschgorin disk is defined as $D_i=\{z\|z-m_{ii}|<\sum_{j=1,j\neq i}^{n}|m_{ij}|\}$, $z\in C$, and the Gerschgorin domain is defined as the union of all the Gerschgorin disks $D_M=\cup_{i=1}^{n}D_i$. All the eigenvalues of matrix M lie in the Gerschgorin domain D_M.

Assumption 4: The PJM $\boldsymbol{\Phi}(k)$ is a diagonally dominant matrix in the following condition, i.e. $|\hat{\varphi}_{ij}(k)|\le b_1$, $b_2\le|\hat{\varphi}_{ii}(k)|\le\gamma b_2$, $\gamma\ge1$, $b_2>b_1(2\gamma+1)(m-1)$, $i=1,\ldots,m$ $j=1,\ldots,2$, $i\neq j$, and all signs of the elements in $\boldsymbol{\Phi}(k)$ are fixed.

Theorem 3: If non-linear system (5) satisfying Assumptions 1–4 is controlled by the EMFAC-TD scheme (21)–(26), for a regulation problem $\boldsymbol{y}^*(k+1)=\boldsymbol{y}^*$, it may be possible to achieve a constant $\lambda_0>0$, such that the following two properties hold for any $\lambda\ge\lambda_0$:

a. The tracking error sequence is convergent, i.e.

$$\lim_{k\to\infty}\|\ \boldsymbol{y}^*-\boldsymbol{y}(k+1)\ \|=0. \tag{27}$$

b. The closed-loop system is BIBO stable, i.e. $\{\boldsymbol{y}(k)\}$ and $\{\boldsymbol{u}(k)\}$ are bounded sequences.

Proof: See Appendix 3. □

Compared with existing model-based control methods, e.g. the MPC [4], the controller of EMFAC-TD is designed in a data-driven

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Table 1 Descriptions of three different controllers

EMFAC-TD		PID	
$\begin{bmatrix}\hat{\varphi}_{11}(0) & \hat{\varphi}_{12}(0)\\ \hat{\varphi}_{21}(0) & \hat{\varphi}_{22}(0)\end{bmatrix}$	$\begin{bmatrix}0.5 & 0.5\\ 0.5 & 0.5\end{bmatrix}$	Controller 1 (longitudinal controller)	
η	1.0	k_p^1	4.5
ρ	1.2	k_I^1	3.6
λ	20	k_d^1	1.1
μ	0.8		
h_{11}	5.0	Controller 2 (lateral controller)	
h_{21}	5.0	k_p^2	12.3
h_{12}	4.0	k_I^2	6.2
h_{22}	4.0	k_d^2	2.6

Table 2 Descriptions of four different tasks

	Driving path	*v*, km/h
Task 1	linear path	[0, 3]
Task 2	circular path	3
Task 3	large curvature curve	[0, 5]
Task 4	arbitrary curve	[0, 5]

manner and only based on the input and output data of the excavator using a new dynamic linearisation technique with a time-varying parameter named the pseudo Jacobian matrix (PJM). Meanwhile, the stability of EMFAC-TD is also verified via rigorous mathematical analysis. Moreover, compared with existing MFAC-based control methods for MIMO systems [35], the proposed EMFAC-TD method also considers the time delay of the system by using the Smith estimating method and the tracking differentiators, such that the existing system time delay can be effectively tackled.

Remark 6: There also exist other direct controller design method, e.g. Wang *et al.* [23] propose a directly DDC method-based subspace method. The difference between the Bontempi *et al.* [18] and the propsed EMFAC-TD can be concluded as follows. First, the model structure information is implicitly involved in the controller design process, since the investigated system is assumed as a linear discrete time-invariant state-space system and the corresponding model parameters are then identified via a series of methods. In the proposed EMFAC-TD method, the controller is directly designed and the original model structure information is not utilised. Moreover, in [18], the controller is designed similarly to the MPC method, and the mainly difference is that the predictor is obtained via the subspace identification algorithm. In the proposed EMFAC-TD method, the controller is designed based on the dynamic linearisations and the controller is derived by minimising the weighted one-step-ahead cost function concerning the tracking error and the control inputs. In addition, compared with Bontempi *et al.* [18], the EMFAC-TD also considers existing time delay and deal with it via the Smith estimated method and tracking differentiators.

Remark 7: In practice, there also exist some special cases, in which severe system dynamics uncertainties and external disturbance exist, i.e. the dramatic change of dynamic loads, etc. In this case, the dynamical linearisation model can be further described as follows:

$$\boldsymbol{y_m}(k+\tau+1) = \boldsymbol{y}(k+\tau) + \boldsymbol{\Phi}(k)\Delta\boldsymbol{u}(k) + \boldsymbol{d}(k), \tag{28}$$

where $\boldsymbol{d}(k) = [d_1(k),\ d_2(k)]^{\mathrm{T}}$ denotes the uncertainties of the system dynamic and external disturbance at time instant k. Then, during the controller design process, the $\boldsymbol{y}(k+\tau+1)$ is replied with $\boldsymbol{y}_m(k+\tau+1)$. There exist references for MFAC method considering the single-input and single-output (SISO) system with the uncertainties of the system dynamic and external disturbance [37, 38], and the corresponding methods for dealing with uncertainties disturbances can be extended to the MIMO systems with dynamic linearisation model (28). More details can be seen in [37, 38].

4 Experimental results

To verify the applicability of the proposed algorithm, experiments are implemented on a practical self-driving excavator in this paper.

The experimental site is in Xuzhou, Jiangsu Province. The programming environment is Codesys, and the data collection, data process, and data analysis and saving are performed using DEWESoft V7.0. The sampling period is 0.05 s.

From Section 2.2, we know that the dynamics of the excavator system can be described in the form of (5). The time delay τ is test to 45 sampling periods here. For comparison, the conventional PID method is applied, which includes two PID controllers for the control of the lateral and longitudinal dynamics, respectively. The controllers are described as follows:

$$y_1 = k_p^1 e^1(k) + k_I^1 \sum_{i=1}^{k} e^1(k) + k_d^1(e^1(k) - e^1(k-1)), \tag{29}$$

$$y_2 = k_p^2 e^2(k) + k_I^2 \sum_{i=1}^{k} e^2(k) + k_d^2(e^2(k) - e^2(k-1)), \tag{30}$$

where k_p^1, k_I^1, and k_d^1 are the parameters of the longitudinal controller and k_p^2, k_I^2, and k_d^2 are the parameters of the lateral controller. $e^1(k)$ is speed error between the desired value and output value at time instant k, described as $y_1^*(k) - y_1(k)$. Moreover, $e^2(k)$ is the error of the pre-aim-yaw angle at time instant k, described as $y_2^*(k) - y_2(k)$. The controller parameters of the two methods are shown in Table 1.

In the sequel, we investigate the tracking results of the excavator. Specifically, the excavator conducts the four tasks shown in Table 2.

4.1 Linear path (Task 1)

To clearly observe the control performance of the proposed control of the speed, the excavator will perform a linear driving task. The test time is 45 s, and the test length is 300 m. The desired speed is described as follows.

$$y_1^* = \begin{cases} 0.1t,\ 0 \le t \le 14 \\ 1.4\,\text{km/h},\ 14 \le t \le 33 \\ 0.9\,\text{km/h},\ 33 \le t \le 45 \end{cases}. \tag{31}$$

The excavator conducts two tests to track the same linear path using the PID control method and the EMFAC-TD method, respectively. Consequently, the control performance for the linear path tracking using the EMFAC-TD and conventional PID method is shown in Fig. 8. It can be seen from Fig. 8 that both methods can accomplish the linear path tracking task. Compared with the conventional PID method, the tracking accuracy using EMFAC-TD is higher, since the result is closer to the desired linear path. During the driving process, the speed control performance for the two tests using PID and EMFAC-TD is shown in Fig. 9.

It can be seen from Fig. 9 that the proposed EMFAC-TD can track the desired speed more accurately than the conventional PID method, e.g. in the time interval [15 s, 30 s]. In addition, when the desired speed changes dramatically at about the 34th second, the EMFAC-TD can still track the desired speed well, whereas the conventional PID method cannot quickly track the desired signal and has oscillations in the time interval [40 s, 45 s].

Moreover, the changes of the estimation for each element of PJM are shown in Fig. 10. It can be seen from Fig. 10 that each element in the PJM is updated during the control process. The results further verify the adaptivity of proposed EMFAC-TD during the control process.

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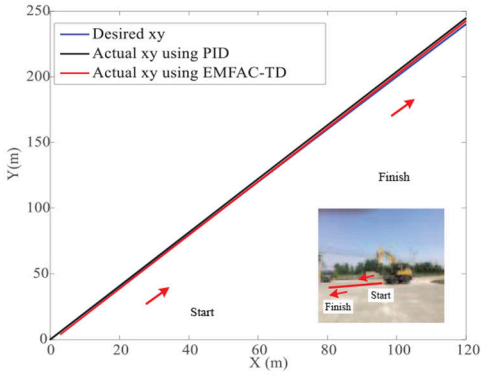


Fig. 8 *Control performance of the path tracking result for the linear look-ahead trajectory*

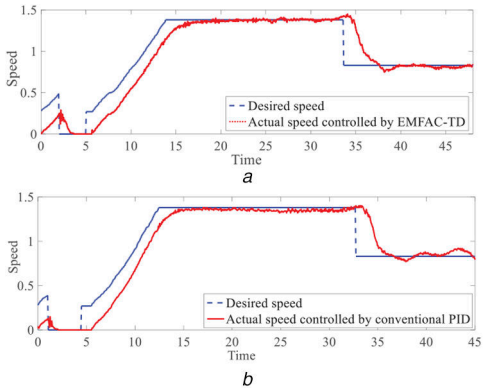


Fig. 9 *Control performance of the speed tracking results for the linear look-ahead trajectory*
(a) EMFAC-TD, ***(b)*** PID

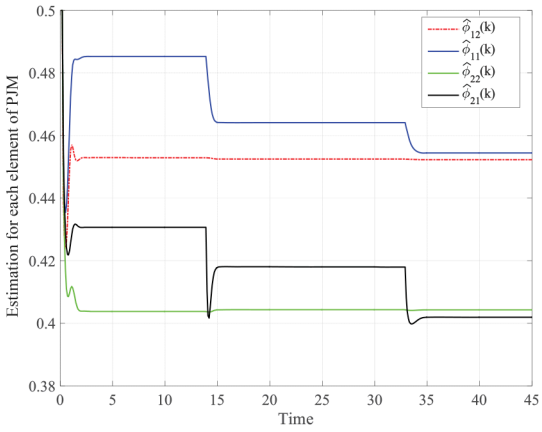


Fig. 10 *Changes of the estimation for each element of PJM for linear task of the excavator*
(a) EMFAC-TD, ***(b)*** PID

4.2 Circular reference path (Task 2)

In most cases, the excavator should search for an object along a specific path. For this reason, we make a circular reference path and let the excavator track this circular reference path in this subsection. The control performances of the tracking path using the proposed method and the conventional PID method are shown in Fig. 11. The driving speed is maintained as 3 km/h.

In Fig. 11, it can be seen that the EMFAC-TD can track the reference path with little error, whereas the conventional PID method has a larger tracking error. The results also verify that when the driving speed of the excavator is fixed, the proposed methods have better performance for the control of the lateral dynamics.

Remark 8: The system output is the driving speed y_1 and the pre-aim-yaw angle y_2. In Theorem 1 and Figs. 5 and 6, we know that the if the path tracking accuracy is high, then the pre-aim-yaw angle y_2 will also track well. To directly observe the performance of the lateral dynamics of the excavator, the path tracking accuracy is utilised to replace the pre-aim-yaw angle y_2 tracking accuracy.

4.3 Large curvature reference path (Task 3)

For some special situations, the excavator needs to make a large curvature turn, e.g. when the excavator meets a sudden obstruction or meets the large curvature of a driving road. For this reason, the excavator conducts a large curvature reference path in this subsection. The control performance is shown in Fig. 12.

From Fig. 12, it can be seen that the EMFAC-TD can adopt changes in the driving speed more quickly than the conventional PID controller when the excavator makes a large curvature turn. The specific numerical error terms of the path tracking accuracy during the driving are is shown in Table 3. The corresponding terms are defined as follows:

$$\mathrm{MSE} = \sum_{k=1}^{N} \| (x_k, y_k) - (x_k^*, y_k^*) \|^2 / N,$$

$$\mathrm{e}_{\max} = \max_k \{(x_k, y_k) - (x_k^*, y_k^*)\},$$

where N is the total number of time instants in one experiment. The results of Table 3 also verify that the path tracking accuracy for this large curvature using EMFAC-TD is higher than that of PID.

For the control performance of the driving speed, to reduce the influence of the inertance of the excavator for the turn during the driving process, the driving speed y_1 is reduced to 3 km/h when the excavator makes a turn, which can be described as

$$y_1^* = \begin{cases} 5\ \text{km/h, linear line} \\ 3\ \text{km/h, sharp curve} \end{cases}. \tag{32}$$

Further, the switch of the desired speed from 5 to 3 km/h during the driving process can be justified according to the following condition:

$$\frac{1}{n} \sum_{i=k+1}^{k+n} C(k) \geq C_0, \tag{33}$$

where $C(k)$ is the curvature of the desired look-ahead trajectory at time instant k and C_0 is a given positive constant. Thus, the equation can be utilised to judge whether the trajectory at time instant $k+1$ is the 'linear line' and decide whether the speed can be switched. The practical significance of (30) is that the excavator should reduce its speed to ensure the safety of the excavator when the average curvature of the future n time instant is large enough (larger than C_0).

The accuracy terms of the driving speed during the driving process are shown in Table 4. The corresponding terms are defined as follows:

$$\mathrm{MS\{E\}}' = \sum_{k=1}^{N} \| (y_1^* - y_1(k)) \|^2 / N,$$

$$e'_{\max} = \max_k \{(y_1^* - y_1(k))\},$$

where N is the total number of time instants in one experiment. From Table 4, it can be seen that both the MS{E}' and the $\{\{e\}\}'_{\max}$ of the proposed method are smaller than those of the conventional PID method. The MSE using proposed method is 0.12 m, reduced nearly 40% than that of conventional PID.

Remark 9: The reasons that the 40% of decreasing the control error can be concluded as the following two aspects. One is that the preview-deviation-yaw-based method is proposed, by which the path tracking problem concerning the horizontal and vertical

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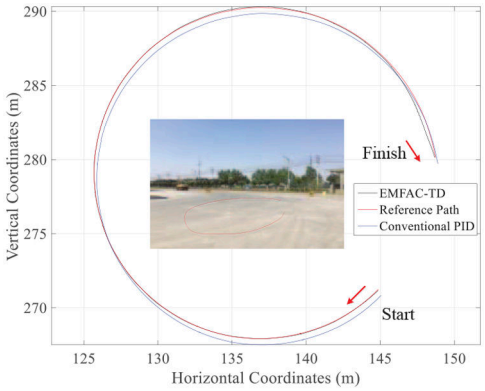


Fig. 11 *Control performance of the path tracking result for the circular look-ahead trajectory*

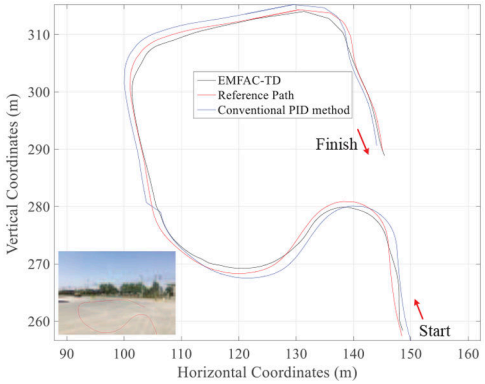


Fig. 12 *Control performance of the path tracking result for the large curvature look-ahead trajectory*

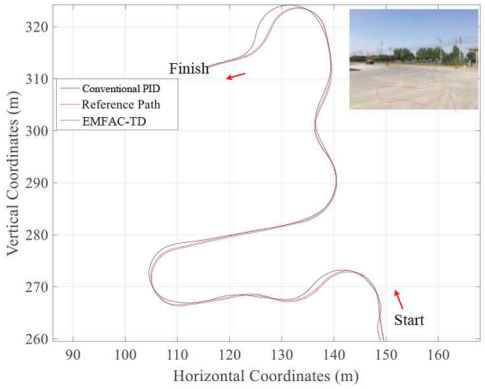


Fig. 13 *Control performance of the path tracking result for arbitrary curve*

coordinates of the excavator is converted into a regulation problem concerning a proposed preview-deviation-yaw, such that the control objective become more clearly and can be controlled more accurate than the previous existing control objective. Another reason is that the proposed control method considers the time delay of the excavator, and the control parameters are online updated according to the practical conditions, such that the control algorithm is adaptive and the control performance is improved than the existing PID control method.

The results of Fig. 12 and Tables 3 and 4 verify that for a large curvature reference path, EMFAC-TD can ensure not only a high path tracking accuracy but also that the driving speed has a small tracking error.

4.4 Arbitrary curve (Task 4)

In most cases, the excavator cannot know the path in advance. Therefore, in this task, we make an arbitrary reference path without any geometrical characteristics to test the control performance of

Table 3 MSE and the maximum error of the path tracking

	MSE, m	e_{max}, m
PID	0.35	1.2
EMFAC-TD	0.12	0.7

Table 4 MSE and the maximum error of the driving speed during the path tracking process

	MSE', m/s	e'_{max}, m/s
PID	0.35	0.75
EMFAC-TD	0.13	0.31

the proposed method. The tracking results using the proposed method and the conventional PID method are shown in Fig. 13.

From Fig. 13, it can be seen that the proposed method (blue curve) tracks the reference path more accurately than the conventional PID method. For example, in the vertical coordinate interval [310 m, 320 m], the path using the conventional PID method has a larger error than that using EMFAC-TD. The reason is that when the excavator makes a large curve turn, the PID controller cannot react to the dramatic changes due to system time delay and the inertance of the excavator, whereas the controller using EMFAC-TD can adapt to sudden changes in the reference path, since the controller parameters can be updated to adapt to these changes.

Remark 10: During the driving process of the excavator, it is inevitable to meet the obstacle for some special situations. For this situation, the local trajectory planning algorithm will be triggered. The local trajectory planning algorithm is applied in the road within a distance of 200 m. When the obstacle is detected, a cluster of feasible paths can be obtained by means of longitudinal and lateral translation for the original desired path without obstacle. Consequently, the new desired path can be determined and the excavator will be controlled to track the new path. The process of local path planning can be shown in Fig. 14, in which a cluster of 16 feasible paths is obtained and anyone in the feasible paths can be selected as the desired path to avoid the obstacle.

Remark 11: Since the experiments are implemented on a practical excavator out of doors, the data collected from the sensors contain the noise and the external disturbance itself. Meanwhile, the experimental results described in Figs. 8–13 using the proposed EMFAC-TD method also show satisfactory performance. Thus, the proposed EMFAC-TD method is insensitive to noise under the normal level. The reason can be concluded as follows. The EMFAC-TD controller is constructed only based on an equivalent time-varying linearisation data model, in which neither the system model is utilised explicitly (or implicitly), nor the unmodelled dynamics is involved. Control performance of EMFAC-TD would not be influenced by traditional unmodelled dynamics affected by noise or external disturbance. Based on this observation, it can be concluded that EMFAC-TD scheme have strong robustness. Besides, it is also worth pointing out that the EMFAC-TD scheme not only has simple and practical structure, but also guarantees BIBO stability and monotonic convergence of the output tracking error for the controlled plant with time delay.

5 Conclusion

In this work, we investigate the path tracking control problem of a self-driving excavator. The dynamics of the self-driving excavator is a typical MIMO system with a time delay, and there exists a strong coupling between the longitudinal driving speed and the lateral wheel angle. For this problem, an EMFAC-TD is proposed. Specifically, the controller is designed via a novel dynamical linearisation technique with a time-varying parameter named the pseudo Jacobian matrix (PJM), and the method also considers the time delay of the system by using the Smith estimating method and tracking differentiators. The main feature of the proposed EMFAC-TD is that the controller is designed in a data-driven manner and is

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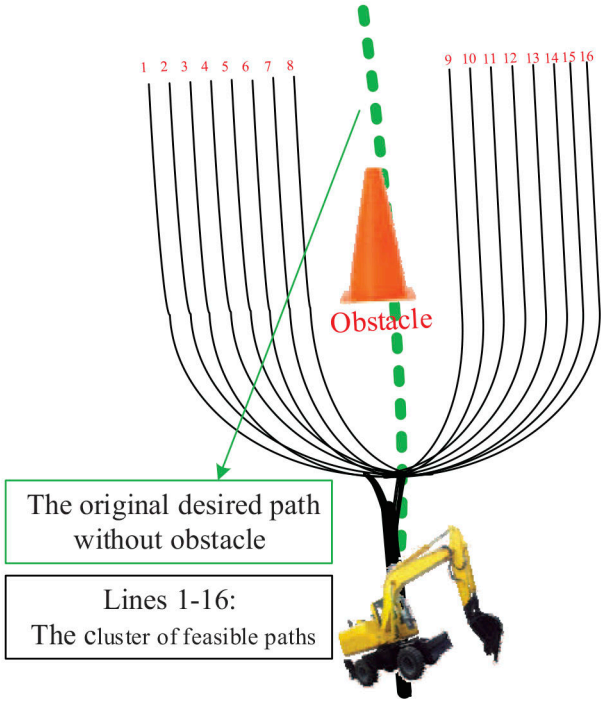


Fig. 14 *Process of local path planning for the situation of obstacle avoidance*

only based on the input–output data of the excavator using a new dynamic linearisation technique, i.e. no explicit knowledge of the plant dynamics is required even for the MIMO system. Further, the stability of the proposed EMFAC-TD is proved via rigorous mathematical analysis. In addition, the validity of the control framework was shown through real-time experiments on a real industrial self-driving excavator, and the corresponding experimental results show that the maximum tracking error of the excavator controlled via EMFAC-TD method can be limited as 0.3–0.7 m for different driving tasks, satisfying the practical requirement, which verify that EMFAC-TD method can improve the tracking performance and complete the required task successfully.

In the future, we will focus on the excavation part of the excavator such that the self-driving excavator will be able to realise automatic driving and mining.

6 Acknowledgments

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8 Appendix

8.1 Appendix 1

Proof of Theorem 2.1: Proof: The entire proof is divided as follows:

Step 1: Proving that

$$\theta \to 0 \Rightarrow (x, y) \to (x^*, y^*). \tag{34}$$

To clearly analyse the lateral dynamics of the excavator, a schematic diagram of the lateral dynamics from the kth to (k+1)th time instants is shown in Fig. 15.

In Fig. 15, $AD = \angle ACB$ (vertical angles). Meanwhile, in ΔAO_1C, $\sin\theta = AE/AO_1$.

Since $\theta \to 0$ and $AO_1 \geq d_{\min} \neq 0$ (as mentioned in (31)), it can be inferred that $AE \to 0$, i.e. $LD \to 0$.

Moreover, because $\sin\angle ACO_1 = AE/AC$ and $AE \to 0$, it can be inferred that $AD \to 0$.

Moreover, since the excavator drives in the direction of the look-ahead point, in ΔO_1O_3C, $X(k+1) < X(k)$. Therefore, it can be inferred that

$$\begin{aligned} &\| (x(k), y(k)) - (x^*, y^*) \|_2 \\ &< \| (x(k+1), y(k+1)) - (x^*, y^*) \|_2. \end{aligned}$$

Define the series as $\{m_k\}$, $m_k = \| (x(k), y(k)) - (x^*, y^*) \|_2$, and since $m_k > 0$ and $\{m_k\}$ is monotonically decreasing, it can be inferred that $\lim_{k\to\infty} m_k = 0$, i.e. $(x, y) \to (x^*, y^*)$.

Step 2: Proving that

$$\theta \to 0 \Leftarrow (x, y) \to (x^*, y^*). \tag{35}$$

Since $(x, y) \to (x^*, y^*)$, it can be obtained that $X(k) \to 0$.

Since $X(k) > AE(k) \geq 0$, it can be inferred that

$$AE(k) \to 0, \tag{36}$$

where $AE(k)$ denotes the lateral distance at time instant k.

Moreover, in ΔAEC, $\sin\angle ACE(k) = AE(k)/AC(k)$. From (36), $\sin\angle ACE(k) \to 0$, i.e. $\angle ACE(k) \to 0$.

In addition, in ΔAEO_1, $\sin\theta(k) = AE(k)/O_1A$, and from (36), it can be inferred that $\theta(k) \to 0$. Thus, conclusion (35) is obtained.□

8.2 Appendix 2

To extend the proof to a more general case, the proof extends the dimension of the control input and control output from 2 to m.

Proof of Theorem 3.1: From (5)

$$\begin{aligned} &\Delta \boldsymbol{y}(k+1) = \\ &f(\boldsymbol{y}(k), \ldots, \boldsymbol{y}(k-n_y), \boldsymbol{u}(k-\tau), \ldots, \boldsymbol{u}(k-\tau-n_u)) \\ &-f(\boldsymbol{y}(k), \ldots, \boldsymbol{y}(k-n_y), \\ &\boldsymbol{u}(k-\tau-1), \boldsymbol{u}(k-\tau-1), \ldots, \boldsymbol{u}(k-\tau-n_u)) + f(\boldsymbol{y}(k), \ldots, \\ &\boldsymbol{y}(k-n_y), \boldsymbol{u}(k-\tau-1), \boldsymbol{u}(k-\tau-1), \ldots, \boldsymbol{u}(k-\tau-n_u)) \\ &-f(\boldsymbol{y}(k-1), \ldots, \boldsymbol{y}(k-n_y-1), \\ &\boldsymbol{u}(k-\tau-1), \ldots, \boldsymbol{u}(k-\tau-n_u-1)). \end{aligned} \tag{37}$$

Let

$$\begin{aligned} \Psi(k) = &f(\boldsymbol{y}(k), \ldots, \boldsymbol{y}(k-n_y), \boldsymbol{u}(k-\tau-1), \\ &\boldsymbol{u}(k-\tau-1), \ldots, \boldsymbol{u}(k-\tau-n_u)) - f(\boldsymbol{y}(k-1), \ldots, \\ &\boldsymbol{y}(k-n_y-1), \boldsymbol{u}(k-\tau-1), \ldots, \boldsymbol{u}(k-\tau-n_u-1)). \end{aligned} \tag{38}$$

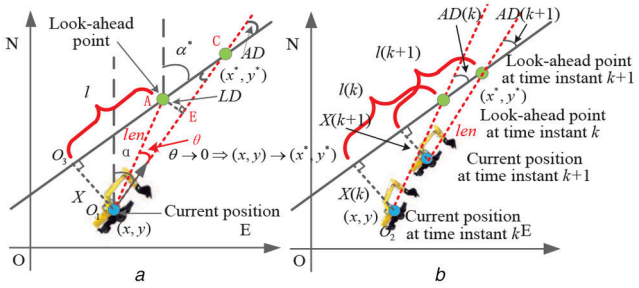


Fig. 15 *Schematic diagram of the lateral dynamics from the kth to (k+1)th time instants*

(a) The schematic diagram of the lateral dynamics at kth time instant, ***(b)*** The schematic diagram of the lateral dynamics at (k+1)th time instant

By virtue of Assumption 1 and the differential mean value theorem, (37) can be rewritten as

$$\Delta \boldsymbol{y}(k+1) = \frac{\partial \boldsymbol{f}^*}{\partial \boldsymbol{u}(k-\tau)} \Delta \boldsymbol{u}(k-\tau) + \Psi(k), \tag{39}$$

where

$$\frac{\partial \boldsymbol{f}^*}{\partial \boldsymbol{u}(k-\tau)} = \begin{bmatrix} \frac{\partial f_1^*}{\partial u_1(k-\tau)} & \frac{\partial f_1^*}{\partial u_2(k-\tau)} & \cdots & \frac{\partial f_1^*}{\partial u_m(k-\tau)} \\ \frac{\partial f_2^*}{\partial u_1(k-\tau)} & \frac{\partial f_2^*}{\partial u_2(k-\tau)} & \cdots & \frac{\partial f_2^*}{\partial u_m(k-\tau)} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m^*}{\partial u_1(k-\tau)} & \frac{\partial f_m^*}{\partial u_2(k-\tau)} & \cdots & \frac{\partial f_m^*}{\partial u_m(k-\tau)} \end{bmatrix},$$

in which $(\partial f_i^*)/(\partial u_j(k))$ represents the partial derivative value of f_i with respect to input u_j at some point in the interval $[u_j(k), u_j(k-1)]$.

For each fixed k, we consider the following equation with the numerical matrix $\eta(k)$

$$\boldsymbol{\Psi}(k) = \boldsymbol{\eta}(k) \Delta \boldsymbol{u}(k-\tau). \tag{40}$$

Since condition $\| \Delta \boldsymbol{u}(k-\tau) \| \neq 0$ holds, (40) must have at least one solution $\boldsymbol{\eta}^*(k)$.

Let $\boldsymbol{\Phi}(k-\tau) = \boldsymbol{\eta}^*(k) + (\partial \mathbf{f}^*/\partial \boldsymbol{u}(k-\tau))$, and then $\Delta \boldsymbol{y}(k+1) = \boldsymbol{\Phi}(k-\tau) \Delta \boldsymbol{u}(k-\tau)$. Moreover, $\| \boldsymbol{\Phi}(k) \| \leq b$ is the direct conclusion of Assumption 2. □

8.3 Appendix 3

Proof of Theorem 3.2: To extend the proof to a more general case, the proof extends the dimension of the control input and control output from 2 to m. The proof includes two steps as follows.

(1) Step 1: Prove the boundedness of the estimation of PJM

Let $\hat{\boldsymbol{\Phi}}(k) = [\hat{\boldsymbol{\phi}}_1^T(k), \ldots, \hat{\boldsymbol{\phi}}_m^T(k)]^T$,

$$\hat{\boldsymbol{\phi}}_i(k) = [\hat{\boldsymbol{\phi}}_{i1}(k), \ldots, \hat{\boldsymbol{\phi}}_{im}(k)], \quad i = 1, 2, \ldots, m$$

The PJM estimation algorithm (21) can be rewritten as

$$\begin{aligned} &\hat{\boldsymbol{\phi}}_i(k) = \hat{\boldsymbol{\phi}}_i(k-\tau-1) \\ &+ \frac{\eta(\Delta \boldsymbol{y}_i(k) - \hat{\boldsymbol{\phi}}_i(k-\tau-1) \Delta \boldsymbol{u}(k-\tau-1)) \Delta \boldsymbol{u}^T(k-\tau-1)}{\mu + \| \Delta \boldsymbol{u}(k-\tau-1) \|^2}, \end{aligned} \tag{41}$$

where $\Delta \boldsymbol{y}_i(k) = \boldsymbol{\phi}_i(k-\tau-1) \Delta \boldsymbol{u}(k-\tau-1), i = 1, 2, \ldots, m$.

Let $\tilde{\boldsymbol{\phi}}_i(k) = \hat{\boldsymbol{\phi}}_i(k) - \boldsymbol{\phi}_i(k)$, and subtracting $\boldsymbol{\phi}_i(k)$ from both sides of (41), it has

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$$\tilde{\boldsymbol{\phi}}_i(k)=\tilde{\boldsymbol{\phi}}_i(k-\tau-1)\left[I-\frac{\eta\Delta\boldsymbol{u}(k-\tau-1)\Delta\boldsymbol{u}^T(k-\tau-1)}{\mu+\|\Delta\boldsymbol{u}(k-\tau-1)\|^2}\right]+\boldsymbol{\phi}_i(k-\tau-1)-\boldsymbol{\phi}_i(k). \tag{42}$$

From Theorem 2, it can be seen that $\|\boldsymbol{\Phi}(k)\|$ has an upper bound, i.e. there exists a positive constant $\bar{c}$ such that $\|\boldsymbol{\Phi}(k)\|\le\bar{b}$, and thus it can be inferred that

$$\|\boldsymbol{\phi}_i(\boldsymbol{k-\tau-1})-\boldsymbol{\phi}_i(\boldsymbol{k})\|\le 2\bar{b}. \tag{43}$$

Taking the norm to (43)

$$\begin{aligned}&\|\tilde{\boldsymbol{\phi}}_i(k)\|\\ &\le\|\tilde{\boldsymbol{\phi}}_i(k-\tau-1)\left[I-\frac{\eta\Delta\boldsymbol{u}(k-\tau-1)\Delta\boldsymbol{u}^T(k-\tau-1)}{\mu+\|\Delta\boldsymbol{u}(k-\tau-1)\|^2}\right]\|\\ &\quad+\|\boldsymbol{\phi}_i(\boldsymbol{k-\tau-1})-\boldsymbol{\phi}_i(\boldsymbol{k})\|\\ &\le\|\tilde{\boldsymbol{\phi}}_i(k-\tau-1)\left[I-\frac{\eta\Delta\boldsymbol{u}(k-\tau-1)\Delta\boldsymbol{u}^T(k-\tau-1)}{\mu+\|\Delta\boldsymbol{u}(k-\tau-1)\|^2}\right]\|+2\bar{b}\end{aligned} \tag{44}$$

For the first term on the right hand of (44)

$$\begin{aligned}&\|\tilde{\boldsymbol{\phi}}_i(k-\tau-1)\left[I-\frac{\eta\Delta\boldsymbol{u}(k-\tau-1)\Delta\boldsymbol{u}^T(k-\tau-1)}{\mu+\|\Delta\boldsymbol{u}(k-\tau-1)\|^2}\right]\|^2\\ &=\|\tilde{\boldsymbol{\phi}}_i(k-\tau-1)\|^2+\left[-2+\frac{\eta\|\Delta\boldsymbol{u}(k-\tau-1)\|^2}{\mu+\|\Delta\boldsymbol{u}(k-\tau-1)\|^2}\right]\\ &\quad\cdot\frac{\eta\|\boldsymbol{\phi}_i(\boldsymbol{k-\tau-1})\Delta\boldsymbol{u}(\boldsymbol{k-\tau-1})\|^2}{\mu+\|\Delta\boldsymbol{u}(k-\tau-1)\|^2}.\end{aligned} \tag{45}$$

Because $0<\eta\le 2$ and $\mu>0$,

$$-2+\frac{\eta\|\Delta\boldsymbol{u}(k-\tau-1)\|^2}{\mu+\|\Delta\boldsymbol{u}(k-\tau-1)\|^2}<0. \tag{46}$$

Thus, it can be inferred that there exists $0<d_1<1$, such that

$$\begin{aligned}&\|\tilde{\boldsymbol{\phi}}_i(k-\tau-1)\left[I-\frac{\eta\Delta\boldsymbol{u}(k-\tau-1)\Delta\boldsymbol{u}^T(k-\tau-1)}{\mu+\|\Delta\boldsymbol{u}(k-\tau-1)\|^2}\right]\|\cdots\\ &\le d_1\|\tilde{\boldsymbol{\phi}}_i(k-\tau-1)\|\end{aligned} \tag{47}$$

Substitute (47) into (45)

$$\begin{aligned}\|\tilde{\boldsymbol{\phi}}_i(k)\|&\le d_1\|\tilde{\boldsymbol{\phi}}_i(k-\tau-1)\|+2\bar{b}\\ &\le\cdots\le d_1^{k-1}\|\tilde{\boldsymbol{\phi}}_i(1-k\tau)\|+\frac{2\bar{b}}{1-d_1},\end{aligned} \tag{48}$$

which means that $\tilde{\boldsymbol{\phi}}_i(k)$ is bounded. Meanwhile, from Theorem 2, $\boldsymbol{\phi}_i(k)$ is bounded. Therefore, $\hat{\boldsymbol{\phi}}(k)$ is bounded.

(2) Step 2: Denote the tracking error as

$$\boldsymbol{e}(k+1)=\boldsymbol{y}^*-\boldsymbol{y}(k+1)=\boldsymbol{y}^*-\boldsymbol{y}(k)-\boldsymbol{\Phi}(k-\tau)\Delta\boldsymbol{u}(k-\tau).$$

Substituting (24), $\boldsymbol{e}(k+1)$ can be written as

$$\begin{aligned}\boldsymbol{e}(k+1)&=\boldsymbol{y}^*-\boldsymbol{y}(k)-\boldsymbol{\Phi}(k-\tau)\\ &\quad\cdot\frac{\rho\boldsymbol{\Phi}^T(k-\tau)(\boldsymbol{y}^*(k+1)-\tau\boldsymbol{y}'_{TD}(k-\tau)-\boldsymbol{y}(k-\tau))}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\end{aligned}$$

Then, using (7), $\boldsymbol{e}(k+1)$ can be further written as

$$\begin{aligned}\boldsymbol{e}(k+1)&=\boldsymbol{y}^*-\boldsymbol{y}(k)-\boldsymbol{\Phi}(k-\tau)\\ &\quad\cdot\frac{\rho\boldsymbol{\Phi}^T(k-\tau)(\boldsymbol{y}^*(k+1)-\boldsymbol{y}(k))}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\\ &=\left[I-\frac{\rho\boldsymbol{\Phi}(k-\tau)\hat{\boldsymbol{\Phi}}^T(k-\tau)}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\right]e(k).\end{aligned} \tag{49}$$

Define z as the eigenvalue of the matrix

$$\left[I-\frac{\rho\boldsymbol{\Phi}_c(k-\tau)\hat{\boldsymbol{\Phi}}_c^T(k-\tau)}{\lambda+\|\boldsymbol{\Phi}_c(k-\tau)\|^2}\right].$$

Then, from Lemma 1, it has

$$D_j=\left\{z\left|\begin{array}{l}\left|z-\left|1-\dfrac{\rho\sum_{i=1}^m\phi_{ji}(k-\tau)\hat{\phi}_{ji}(k-\tau)}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\right|\right|\\ \le\displaystyle\sum_{l=1,l\ne j}^m\left|\frac{\rho\sum_{i=1}^m\phi_{ji}(k-\tau)\hat{\phi}_{li}(k-\tau)}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\right|\end{array}\right.\right\}. \tag{50}$$

According to the triangle inequality, (50) can be obtained as

$$D_j=\left\{z\left|\begin{array}{l}|z|\le\left|1-\dfrac{\rho\sum_{i=1}^m\phi_{ji}(k-\tau)\hat{\phi}_{ji}(k-\tau)}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\right|\\ +\displaystyle\sum_{l=1,l\ne j}^m\left|\frac{\rho\sum_{i=1}^m\phi_{ji}(k-\tau)\hat{\phi}_{li}(k-\tau)}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\right|\end{array}\right.\right\}. \tag{51}$$

From Lemma 1 and Assumption 3, it has

$$\begin{aligned}&b_2\le\left|\hat{\phi}_{ii}(k)\right|\le\alpha b_2,\ \left|\hat{\phi}_{ij}(k)\right|\le b_1,\\ &i=1,2,\ldots,m,\quad j=1,2,\ldots,m,i\ne j,\\ &b_2\le|\phi_{ii}(k)|\le\alpha b_2,|\phi_{ij}(k)|\le b_1,\\ &i=1,2,\ldots,m,\quad j=1,2,\ldots,m,i\ne j\end{aligned}$$

Therefore, the following two inequalities hold:

$$\begin{aligned}&\left|1-\frac{\rho\sum_{i=1}^m\phi_{ji}(k-\tau)\hat{\phi}_{ji}(k-\tau)}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\right|\\ &\le 1-\frac{\rho\left|\phi_{jj}(k-\tau)\right|\left|\hat{\phi}_{ji}(k-\tau)\right|}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\\ &\le 1-\frac{\rho b_2^2}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\end{aligned} \tag{52}$$

$$\begin{aligned}&\sum_{l=1,l\ne j}^m\left|\frac{\rho\sum_{i=1}^m\phi_{ji}(k-\tau)\hat{\phi}_{li}(k-\tau)}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\right|\\ &\le\rho\sum_{l=1,l\ne j}^m\frac{\sum_{i=1}^m\left|\phi_{ji}(k-\tau)\right|\left|\hat{\phi}_{li}(k-\tau)\right|}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\\ &=\rho\sum_{l=1,l\ne j}^m\frac{\left|\phi_{jj}(k-\tau)\right|\left|\hat{\phi}_{lj}(k-\tau)\right|}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\\ &\quad+\rho\sum_{l=1,l\ne j}^m\frac{\left|\phi_{jl}(k-\tau)\right|\left|\hat{\phi}_{ll}(k-\tau)\right|}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}\\ &\quad+\rho\sum_{l=1,l\ne j}^m\frac{\sum_{i=1,i\ne j,l}^m\left|\phi_{ji}(k-\tau)\right|\left|\hat{\phi}_{li}(k-\tau)\right|}{\lambda+\|\boldsymbol{\Phi}(k-\tau)\|^2}.\end{aligned} \tag{53}$$

From Assumption 3, $b_2>b_1(2\alpha+1)(m-1)$. Then, summing (52) and (53) produces

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$$
\begin{aligned}
&\left|1-\frac{\rho\sum_{i=1}^{m}\phi_{ji}(k-\tau)\hat{\phi}_{ji}(k-\tau)}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2}\right| \\
&\quad+\sum_{l=1,l\neq j}^{m}\left|\frac{\rho\sum_{i=1}^{m}\phi_{ji}(k-\tau)\hat{\phi}_{li}(k-\tau)}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2}\right| \\
&\le 1-\rho\frac{b_2^2-2\alpha b_2 b_1(m-1)-b_1^2(m-1)(m-2)}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2} \\
&=1-\rho\frac{b_2(b_2-2\alpha b_1(m-1))-b_1^2(m-1)(m-2)}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2} \qquad (54)\\
&<1-\rho\frac{b_2 b_1(m-1)-b_1^2(m-1)(m-2)}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2} \\
&<1-\rho\frac{b_1(m-1)[b_2-b_1(m-1)]}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2} \\
&<1-\rho\frac{2\gamma b_1^2(m-1)^2}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2}.
\end{aligned}
$$

Therefore, there exists a $\lambda_{\min}>0$, such that when $\lambda>\lambda_{\min}$, it has

$$
0<M_1<\frac{2\gamma b_1^2(m-1)^2}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2}<\frac{2\gamma b_1^2(m-1)^2}{\lambda_{\min}+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2}<1. \quad (55)
$$

Thus, it can be inferred that

$$
\begin{aligned}
&\left|1-\frac{\rho\sum_{i=1}^{m}\phi_{ji}(k-\tau)\hat{\phi}_{ji}(k-\tau)}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2}\right| \\
&\quad+\sum_{l=1,l\neq j}^{m}\left|\frac{\rho\sum_{i=1}^{m}\phi_{ji}(k-\tau)\hat{\phi}_{li}(k-\tau)}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2}\right| \qquad (56)\\
&<1-\rho M_1<1.
\end{aligned}
$$

According to Lemma 1 and (56)

$$
s\left[I-\frac{\rho\boldsymbol{\Phi}(k-\tau)\hat{\boldsymbol{\Phi}}^T(k-\tau)}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2}\right]<1-\rho M_1, \quad (57)
$$

where $s[M]$ is defined as the spectral radius of the matrix M. From the conclusion of the spectral radius, there exists an arbitrary small positive constant ε_1 such that

$$
\begin{aligned}
&\|\,I-\frac{\rho\boldsymbol{\Phi}(k-\tau)\hat{\boldsymbol{\Phi}}^T(k-\tau)}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2}\,\|_v \\
&<s\left[I-\frac{\rho\boldsymbol{\Phi}_l(k-\tau)\hat{\boldsymbol{\Phi}}^T(k-\tau)}{\lambda+\|\,\boldsymbol{\Phi}_l(k-\tau)\,\|^2}\right]+\varepsilon_1 \qquad (58)\\
&\le 1-\rho M_1+\varepsilon_1<1,
\end{aligned}
$$

where $\|\,M\,\|_v$ is defined as the compatible norm of matrix M. Let $d_3=1-\rho M_1+\varepsilon_1$, and the norm for (51) can be written as

$$
\begin{aligned}
\|\,\boldsymbol{e}(k+1)\,\|_v &\le \|\,I-\frac{\rho\boldsymbol{\Phi}(k-\tau)\hat{\boldsymbol{\Phi}}^T(k-\tau)}{\lambda+\|\,\boldsymbol{\Phi}(k-\tau)\,\|^2}\,\|_v\,\|\,\boldsymbol{e}(k)\,\|_v \qquad (59)\\
&\le d_3\|\,\boldsymbol{e}(k)\,\|_v\le\cdots\le d_3^k\|\,\boldsymbol{e}(1)\,\|_v.
\end{aligned}
$$

Thus, conclusion (27) is proven.

Moreover, since $\boldsymbol{y}^*$ is a constant, and $\boldsymbol{e}(k)$ is bounded, it can be inferred that $\boldsymbol{y}(k)$ is bounded. In addition, since $\hat{\Phi}_c(k)$ is bounded, there must exist a positive constant C_2 such that

$$
\|\,\rho\boldsymbol{\Phi}^T(k)/(\lambda+\|\,\boldsymbol{\Phi}_l(k-\tau)\,\|^2)\,\|<M_2. \quad (60)
$$

Meanwhile

$$
\begin{aligned}
\|\,u(k)\,\|_v &\le \|\,\Delta\boldsymbol{u}(k)\,\|_v+\|\,\Delta\boldsymbol{u}(k-1)\,\|_v \\
&\quad+\cdots+\|\,\Delta\boldsymbol{u}(1)\,\|_v+\|\,u(0)\,\|_v \\
&=M_2(\|\,\boldsymbol{e(k+\tau)}\,\|_v+\|\,\boldsymbol{e(k+\tau-1)}\,\|_v \qquad (61)\\
&\quad+\cdots+\|\,\boldsymbol{e(1+\tau)}\,\|_v)+\|\,\boldsymbol{u(0)}\,\|_v \\
&<M_2\frac{\|\,\boldsymbol{e(1+\tau)}\,\|_v}{1-d_2}+\|\,\boldsymbol{u(0)}\,\|_v
\end{aligned}
$$

Thus, the input sequence $\{\boldsymbol{u(k)}\}$ is bounded. □

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